Caron 2020 Pediatric ALL — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Malignant manifold is sample-mixed AND continuous. 8 patients across 3 genetic ALL groups (ETV6-RUNX1 Pre-B, high-hyperdiploid B-ALL, Pre-T) merge on ONE connected component after Harmony (θ=10) on sample. iLISI 1.00→4.71 of 8. Leiden split is proliferation (cycling 0.87 vs 0.06), not B-vs-T lineage; 0 islands removed.

Gate 2 — module geography → GO (non-branching)

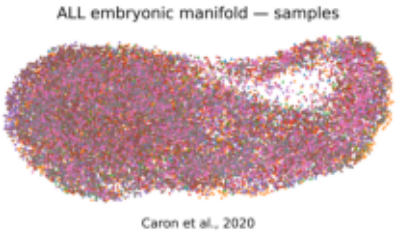
Embryonic pattern present as organized cross-repression, NOT a convergence hub. Gast & pSom sit at opposite poles (smoothed r=−0.45; joint-high enrichment 0.81<1); Crest is an inverse/terminal gradient (pSom–Crest r=−0.72). Root Gast-high (ρ=−0.72 vs pseudotime) → terminus Crest-high (ρ=+0.33). No bifurcation.

Gate 3 — hub cross-repression → NA

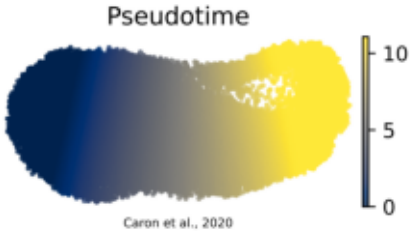
Not evaluated — non-branching topology (principal curve, 2 tips, 0 forks, fold-free). No tree node-clusters, so no mutual Gast–pSom cross-repression test applies. Terminates at the expression step.

STEP FIGURES

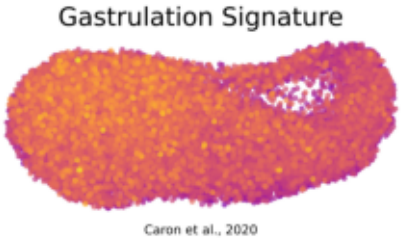
Manifold — samples



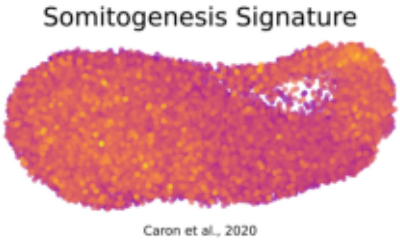
Pseudotime



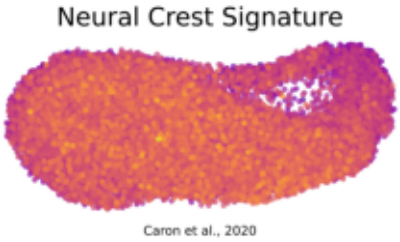
Gastrulation



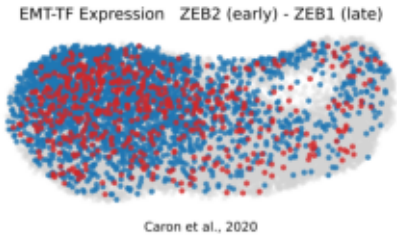
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric acute lymphoblastic leukemia (Caron et al., 2020), 10x — 8 malignant samples / 25,788 cells across 3 genetic ALL groups (ETV6-RUNX1 Pre-B, high-hyperdiploid B-ALL, Pre-T).

Verdict.
Pattern reproduces as the NON-BRANCHING case: a single embryonic axis with organized Gast–pSom cross-repression (r=−0.45) and an inverse terminal neural-crest gradient — a strict linear subset of the Puram hub-and-spoke, with no bifurcation. Root Gast/pSom-high → terminus Crest-high. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (root) → ZEB1 (terminus).

Parameter choices

Min-cell threshold	floor 100 (no dominant cliff; all 8 samples ≥1585) → 8/8 kept, 25,788 cells	**Integration**	Harmony on sample, θ=10 (same-sample nbr frac 0.999→0.272; 3 genetic ALL groups)
Embedding	PCA 50→20 on X_pca_harmony, k=30, spread=1.3, seed=1; radius-graph 0 removed	**Geometry**	principal CURVE (scFates EPG N=20 λ=0.3 μ=100); PPT over-forks a dense blob; curve fold-free
Root	curve tip 0 (Gast/pSom-high, Crest-low) — never Crest (late) or a cycling state	**Modules**	score_genes union control pool (Gast 66/81, pSom 77/98, Crest 93/104); RAW per-cell scores
EMT-TFs (>3%)	ZEB2 (40.6%), ZEB1 (8.9%), SNAI1 (6.8%); overlay pair ZEB2→ZEB1		